

Mobile-fillable review packet

20 Electric Circuits candidates | human release gate

- 1. Download this PDF and open it in Adobe Acrobat Reader or a PDF viewer that saves AcroForm fields.
- 2. For every question, select exactly one PASS or FAIL value for each of the five checks, then one final decision.
- 3. If any check fails, choose REVISE or REJECT and record a precise correction in Notes.
- 4. Save the completed PDF with a new name, then run the included PDF importer. The pipeline remains blocked until its JSON output validates.

Reviewer full name

Qualification / role

Review date (YYYY-MM-DD)

Reviewer email or identifier

Reviewer attestation

Type exactly: I independently reviewed the Batch 002 questions, answers and solutions and approve only the question IDs marked PASS below.

Source lock

BATCH_002_ELECTRIC_CIRCUITS.jsonl | SHA-256 23b749225f9128864b1e1545c865ba1d08b8ee706b3faf69e6202099b913db36

This form records review evidence; it does not itself promote or release questions.

QUESTION

A 6 microfarad capacitor and a 3 microfarad capacitor are connected in series across an 18 V DC source. Determine the total energy stored by this capacitor network in millijoules.

DECLARED ANSWER

0.324

INDEPENDENT RECOMPUTATION EVIDENCE

Ceq=2 microfarads; (1/2)Ceq(18)^2=0.324 mJ.

CANONICAL SOLUTION

The series equivalent capacitance is C_eq=(6 x 3)/(6+3)=2 microfarads. The stored energy is (1/2) C_eq V^2=(1/2)(2 x 10^-6)(18)^2=324 x 10^-6 J=0.324 mJ. Final answer: 0.324

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

Two coupled inductors in a series-aiding network have $L_1=4\text{ H}$, $L_2=9\text{ H}$ and coefficient of coupling $k=0.5$. Determine the equivalent inductance.

OPTIONS

A. 7 H B. 13 H C. 19 H D. 25 H

DECLARED ANSWER

C

INDEPENDENT RECOMPUTATION EVIDENCE

$M=0.5\sqrt{4\times 9}=3\text{ H}$; series-aiding $L_{eq}=4+9+2M=19\text{ H}$, option C.

CANONICAL SOLUTION

The mutual inductance is $M=k\sqrt{L_1 L_2}=0.5\sqrt{4\times 9}=3\text{ H}$. For a series-aiding connection, $L_{eq}=L_1+L_2+2M=4+9+6=19\text{ H}$. Final answer: C

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

At the input port of a linear network, a 4 ohm resistor is connected from the port to ground. A dependent current source of value 0.5 times the resistor current is also directed from the port to ground. Determine the resistance seen at the port.

OPTIONS

A. 2 ohm B. 8/3 ohm C. 4 ohm D. 6 ohm

DECLARED ANSWER

B

INDEPENDENT RECOMPUTATION EVIDENCE

A test voltage v draws $v/4+0.5(v/4)=3v/8$; $v/i=8/3$ ohm, option B.

CANONICAL SOLUTION

Apply a test voltage v at the port. The resistor current is $v/4$ and the dependent-source current is $0.5(v/4)$. Hence the total input current is $i=(3v/8)$, so the driving-point resistance is $v/i=8/3$ ohm. Final answer: B

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

Select every correct statement about ideal passive network elements under finite, impulse-free excitation; one or more options may be correct.

OPTIONS

A. The energy stored in a capacitor is $(1/2)Cv^2$. B. The energy stored in an inductor is Li^2 . C. A capacitor voltage cannot change instantaneously. D. An inductor current cannot change instantaneously.

DECLARED ANSWER

A; C; D

INDEPENDENT RECOMPUTATION EVIDENCE

Ideal stored energies are $Cv^2/2$ and $Li^2/2$; capacitor voltage and inductor current cannot jump under finite impulse-free excitation.

CANONICAL SOLUTION

Capacitor energy is $(1/2)Cv^2$ and inductor energy is $(1/2)Li^2$, so option B misses the factor $1/2$. With finite impulse-free excitation, capacitor voltage and inductor current cannot change instantaneously. Therefore A, C and D are correct. Final answer: A, C, D

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

In a resistive network, node v1 is connected to a 12 V voltage source through 3 ohm, to ground through 6 ohm, and to node v2 through 2 ohm. Node v2 is connected to ground through 4 ohm. Using KCL and node-voltage analysis, determine v2 in volts.

DECLARED ANSWER

4

INDEPENDENT RECOMPUTATION EVIDENCE

KCL gives $v_1=1.5v_2$ and $6v_1-3v_2=24$; hence $v_2=4$ V.

CANONICAL SOLUTION

KCL at v2 gives $(v_2-v_1)/2+v_2/4=0$, hence $v_1=1.5v_2$. KCL at v1 gives $(v_1-12)/3+v_1/6+(v_1-v_2)/2=0$. Multiplying by 6 yields $6v_1-3v_2=24$. Substitution gives $6(1.5v_2)-3v_2=24$, so $v_2=4$ V. Final answer: 4

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

Two non-reference node voltages satisfy $v_1 - v_2 = 5\text{ V}$ because of an ideal voltage source between them. A 2 ohm branch connects v_1 to ground, a 3 ohm branch connects v_2 to ground, and a 4 A current source injects current into v_2 . Using supernode KCL, determine v_1 in volts.

DECLARED ANSWER

6.8

INDEPENDENT RECOMPUTATION EVIDENCE

$v_1/2 + v_2/3 = 4$ with $v_1 - v_2 = 5$ gives $v_2 = 1.8\text{ V}$ and $v_1 = 6.8\text{ V}$.

CANONICAL SOLUTION

KCL for the supernode is $v_1/2 + v_2/3 = 4$. The voltage-source constraint is $v_1 = v_2 + 5$. Substituting and multiplying by 6 gives $3(v_2 + 5) + 2v_2 = 24$, so $v_2 = 1.8\text{ V}$ and $v_1 = 6.8\text{ V}$. Final answer: 6.8

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

Clockwise mesh currents I_1 and I_2 in a resistor network satisfy the KVL equations $5I_1 - 2I_2 = 8$ and $-2I_1 + 4I_2 = 2$. Determine the current in their common branch, directed with I_1 .

OPTIONS

A. 0.625 A B. 1.625 A C. 2.250 A D. 3.875 A

DECLARED ANSWER

A

INDEPENDENT RECOMPUTATION EVIDENCE

The two equations give $I_1 = 2.25$ A and $I_2 = 1.625$ A; $I_1 - I_2 = 0.625$ A, option A.

CANONICAL SOLUTION

Solving the two mesh-current equations gives $I_1 = 36/16 = 2.25$ A and $I_2 = 26/16 = 1.625$ A. The common-branch current in the I_1 direction is $I_1 - I_2 = 0.625$ A. Final answer: A

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

Select every correct statement about node-voltage analysis of a connected electrical network; one or more options may be correct.

OPTIONS

A. One node is selected as the reference node. B. An ideal voltage source between two non-reference nodes can be handled using a supernode. C. For n nodes, at most $n-1$ independent KCL node equations are required. D. The method cannot be used when dependent sources are present.

DECLARED ANSWER

A; B; C

INDEPENDENT RECOMPUTATION EVIDENCE

A reference node, a supernode for an inter-node ideal voltage source, and at most $n-1$ independent KCL equations are valid; dependent sources do not prohibit nodal analysis.

CANONICAL SOLUTION

Node-voltage analysis selects a reference, uses supernodes for ideal voltage sources between unknown nodes, and requires at most $n-1$ independent KCL equations. Dependent sources are permitted when their controlling relations are included. Thus A, B and C are correct. Final answer: A, B, C

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

An 18 V ideal voltage source feeds a 3 ohm series resistor and an output node; a 6 ohm resistor connects that node to ground. A 4 ohm load is then connected from the output node to ground. Determine the load current using the Thevenin equivalent of the source network.

OPTIONS

A. 1 A B. 2 A C. 3 A D. 4 A

DECLARED ANSWER

B

INDEPENDENT RECOMPUTATION EVIDENCE

$V_{th}=18(6/9)=12$ V and $R_{th}=3||6=2$ ohm; $I_L=12/(2+4)=2$ A, option B.

CANONICAL SOLUTION

The open-circuit voltage is $V_{th}=18(6/(3+6))=12$ V. With the independent source deactivated, $R_{th}=3$ parallel 6=2 ohm. Therefore the 4 ohm load current is $12/(2+4)=2$ A. Final answer: B

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A Norton network consists of a 3 A current source in parallel with 6 ohm. When a 3 ohm load is connected across the network, determine the load current.

OPTIONS

A. 0.5 A B. 1 A C. 2 A D. 3 A

DECLARED ANSWER

C

INDEPENDENT RECOMPUTATION EVIDENCE

Current division gives $I_L=3[6/(6+3)]=2$ A, option C.

CANONICAL SOLUTION

By current division, the 3 A Norton current splits between 6 ohm and 3 ohm. The load current is $3[6/(6+3)]=2$ A. Final answer: C

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A sinusoidal network has Thevenin RMS voltage 10 V and Thevenin impedance $3+j4$ ohm. A conjugately matched load is connected. Determine the maximum average load power in watts.

DECLARED ANSWER

8.333

INDEPENDENT RECOMPUTATION EVIDENCE

Conjugate matching gives $P_{max}=|V_{th}|^2/(4R_{th})=100/12=8.333$ W.

CANONICAL SOLUTION

For conjugate matching, the load impedance is $3-j4$ ohm. The maximum average load power is $|V_{th}|^2/(4R_{th})=10^2/(4 \times 3)=8.333$ W. Final answer: 8.333

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

Select every correct statement about superposition in a linear electrical network; one or more options may be correct.

OPTIONS

A. An independent ideal voltage source is replaced by a short circuit when deactivated. B. An independent ideal current source is replaced by an open circuit when deactivated. C. Dependent sources remain active. D. The total power cannot in general be obtained by adding the powers caused by the individual sources.

DECLARED ANSWER

A; B; C; D

INDEPENDENT RECOMPUTATION EVIDENCE

Independent voltage/current sources become shorts/opens, dependent sources remain, and nonlinear power does not superpose.

CANONICAL SOLUTION

Superposition deactivates only independent sources: ideal voltage sources become shorts and ideal current sources become opens. Dependent sources remain because they are part of the network model. Voltages and currents superpose, but power is nonlinear, so the separate-source powers do not generally add. All four statements are correct. Final answer: A, B, C, D

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

In a first-order RC network, the capacitor voltage is initially 2 V and its steady-state value after switching is 10 V. The resistance is 2 kilo-ohm and the capacitance is 100 microfarad. Determine the capacitor voltage, in volts, one time constant after switching.

DECLARED ANSWER

7.057

INDEPENDENT RECOMPUTATION EVIDENCE

$\tau = RC = 0.2 \text{ s}$; $v_C(\tau) = 10 + (2 - 10)e^{-1} = 7.057 \text{ V}$.

CANONICAL SOLUTION

The time constant is $\tau = RC = (2000)(100 \times 10^{-6}) = 0.2 \text{ s}$. At $t = \tau$, $v_C = 10 + (2 - 10)e^{-1} = 10 - 8/e = 7.057 \text{ V}$. Final answer: 7.057

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A 12 V DC source is applied at $t=0$ to an initially unenergized series RL network with resistance 4 ohm and inductance 2 H. Determine the inductor current in amperes at $t=0.5 \ln(2)$ seconds.

DECLARED ANSWER

1.5

INDEPENDENT RECOMPUTATION EVIDENCE

$i_{\infty}=3$ A, $\tau=L/R=0.5$ s and $t/\tau=\ln 2$; $i=3(1-1/2)=1.5$ A.

CANONICAL SOLUTION

The final current is $12/4=3$ A and the time constant is $\tau=L/R=2/4=0.5$ s. Here $t/\tau=\ln(2)$, so $i_L=3[1-e^{(-\ln(2))}]=3(1-1/2)=1.5$ A. Final answer: 1.5

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A series RLC network has inductance 50 mH and capacitance 20 microfarad. Determine its undamped resonant angular frequency.

OPTIONS

A. 100 rad/s B. 500 rad/s C. 1000 rad/s D. 5000 rad/s

DECLARED ANSWER

C

INDEPENDENT RECOMPUTATION EVIDENCE

$LC=10^{-6}$ and $\omega_0=1/\sqrt{LC}=1000$ rad/s, option C.

CANONICAL SOLUTION

The resonant angular frequency is $\omega_0=1/\sqrt{LC}$. Since $LC=(50 \times 10^{-3})(20 \times 10^{-6})=10^{-6}$, $\omega_0=1/10^{-3}=1000$ rad/s. Final answer: C

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A series RLC network has resistance 20 ohm and inductance 0.1 H. Determine its half-power bandwidth in angular frequency.

OPTIONS

A. 100 rad/s B. 200 rad/s C. 500 rad/s D. 2000 rad/s

DECLARED ANSWER

B

INDEPENDENT RECOMPUTATION EVIDENCE

Series-RLC angular bandwidth is $R/L=20/0.1=200$ rad/s, option B.

CANONICAL SOLUTION

For a series RLC resonance, the angular-frequency bandwidth is $\Delta\omega=R/L=20/0.1=200$ rad/s. Final answer: B

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A single-phase impedance load absorbs 8 kW at 0.8 lagging power factor. Select every correct statement about this sinusoidal steady-state network; one or more options may be correct.

OPTIONS

A. The load reactive power is 6 kvar. B. The apparent power magnitude is 10 kVA. C. The load current leads the load voltage. D. A 6 kvar shunt capacitor would correct the source power factor to unity.

DECLARED ANSWER

A; B; D

INDEPENDENT RECOMPUTATION EVIDENCE

$\tan(\arccos 0.8)=0.75$ gives $Q=6$ kvar and $|S|=10$ kVA; the current lags and 6 kvar capacitive compensation reaches unity power factor.

CANONICAL SOLUTION

With $\cos(\phi)=0.8$, $\tan(\phi)=0.75$. Hence the reactive power is $8 \times 0.75=6$ kvar and the apparent-power magnitude is $8/0.8=10$ kVA. A lagging load current does not lead its voltage. A 6 kvar capacitive shunt cancels the inductive reactive power, so A, B and D are correct. Final answer: A, B, D

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A balanced three-phase, star-connected impedance load of $8+j6$ ohm per phase is supplied at a line voltage of 400 V RMS. Determine the total real power in kilowatts.

DECLARED ANSWER

12.8

INDEPENDENT RECOMPUTATION EVIDENCE

$|Z|=10$ ohm and $\cos(\phi)=0.8$; three-phase real power evaluates to 12.8 kW.

CANONICAL SOLUTION

The phase-voltage magnitude is $400/\sqrt{3}$ V, the phase-current magnitude is $[400/\sqrt{3}]/10$ A, and the load power factor is 8/10. Thus the total real power is $\sqrt{3}(400)[400/(\sqrt{3}10)](0.8)=12800$ W=12.8 kW. Final answer: 12.8

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A two-port network has $z_{11}=4\text{ ohm}$, $z_{12}=z_{21}=2\text{ ohm}$ and $z_{22}=5\text{ ohm}$. Port 2 is terminated by a 3 ohm load. Determine the input impedance in ohms.

DECLARED ANSWER

3.5

INDEPENDENT RECOMPUTATION EVIDENCE

$Z_{in}=z_{11}-z_{12}z_{21}/(z_{22}+Z_L)=4-4/8=3.5\text{ ohm}$.

CANONICAL SOLUTION

For a z-parameter two-port terminated by Z_L , the input impedance is $Z_{in}=z_{11}-(z_{12} z_{21})/(z_{22}+Z_L)$. Therefore $Z_{in}=4-(2 \times 2)/(5+3)=4-0.5=3.5\text{ ohm}$. Final answer: 3.5

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)

QUESTION

A balanced three-phase, delta-connected impedance load has 12 ohm magnitude and 30 degree lagging phase angle per branch. It is supplied from a 240 V RMS line. Determine the approximate total real power.

OPTIONS

A. 7.20 kW B. 12.47 kW C. 14.40 kW D. 24.94 kW

DECLARED ANSWER

B

INDEPENDENT RECOMPUTATION EVIDENCE

Delta phase current is $240/12=20$ A; $3(240)(20)\cos30=12.47$ kW, option B.

CANONICAL SOLUTION

For a delta load the phase voltage equals the line voltage, so each phase current is $240/12=20$ A. The total real power is $3(240)(20)\cos(30\text{ degrees})=12470.8$ W, approximately 12.47 kW. Final answer: B

HUMAN DECISION FIELDS

CHECK	PASS	FAIL
Technical correctness	☒	☐
Answer correctness	☒	☐
Solution correctness	☒	☐
Clarity / ambiguity	☒	☐
Originality-conflict check	☒	☐

Decision

☒ PASS

☐ REVISE

☐ REJECT

Notes (required for REVISE or REJECT)